Houssam Jardini

AI / Machine Learning Engineer

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SUMMARY

AI/ML engineer building production systems end to end — training models from scratch and shipping the applications around them. Work spans offline document AI, computer vision, NLP and reinforcement learning, including a table-structure recognition model trained from scratch and a monetized mobile product live on Google Play.

EXPERIENCE

AI / Machine Learning Engineer Intern (PFE) — *Harmony Technology*, Data & AI Unit — Feb 2026 – Aug 2026

- Designed and shipped **DocStruct**, a fully offline document-parsing and structured-extraction platform, meeting every specification target on CPU-only hardware.
- Trained a **9.71M-parameter table-structure recognition model from scratch**, reaching parity with a 53M-parameter state-of-the-art model at a fraction of the compute cost.
- Benchmarked 12 local LLMs, 6 layout-detection stacks and 3 OCR engines on real production documents to drive both systems' design decisions.

Programming & IT Instructor — *Group Academy* (freelance, part-time) — Apr 2026 – Present

- Teach programming, algorithms and IT fundamentals to university students, professionals and engineers.

Founder & CEO — *MotionSync*, junior enterprise via Injaz Al-Maghrib — Mar 2025 – Oct 2025

- Led product, team and pitch for an AI fitness platform delivering real-time pose estimation, form correction and rep counting at 24–28 FPS on mobile.
- Won the **Boeing Innovation Prize** at the Injaz Al-Maghrib National Competition, plus the **Pitch Deck Prize and 3rd Grand Prize** at the SporTech Hackathon, Marrakech.

Data Analyst & AI Engineer Intern — *Arteka*, Amiens, France (remote) — Jul 2024 – Oct 2024

- Engineered a Unity 3D synthetic annotation pipeline generating **10,000+ labelled images**, cutting manual labelling time **98%**; trained YOLO segmentation for 28-class pitch detection.

AI & Robotics Intern — *Code212*, Kénitra — Aug 2023

- Built a vision-controlled robotic gripper from real-time hand-landmark detection (MediaPipe + OpenCV at 30 FPS, Arduino), teleoperated live by hand gestures in the university robotics lab.

PROJECTS

DocStruct — offline document parsing & structured extraction — 2026

- Converts PDF, Office and image formats into Markdown, JSON and user-defined schemas with **no cloud API call at any stage**, CPU-only on 8 GB RAM.
- Custom PaddleOCR engine in the Docling pipeline plus a local GGUF LLM module whose **schema-constrained decoding raised field accuracy +6.5 pts and JSON validity to 99%**; React 19 workspace over a hardened FastAPI backend.

Table-Structure Recognition Model — research contribution — 2026

- Image→OTSL→HTML model (PP-LCNet encoder, transformer decoder, multi-scale fusion, auxiliary bbox head dropped at inference) scoring **0.872 S-TEDS on real scanned tables — within 0.016 of TableFormer at 5.5× fewer parameters**.
- Selected across 12 controlled generations on an **out-of-distribution benchmark** rather than the in-domain standard; grammar-constrained decoding prevents runaway generation by construction. Trained 120k steps on 1× NVIDIA L4 over a ~1.2M-row loader.

MasterCV — AI job-application platform · *live on Google Play (closed testing)* — 2026

- Solo-built Expo / React Native app that researches a target company, tailors CV and cover letter, and exports ATS-safe PDFs — **14 Supabase Edge Functions**, per-user row-level security, server-side-only LLM access, RevenueCat monetization.

VoxMea — real-time sign language translation — 2025

- **88K-parameter MicroTransformer** at **96.23% accuracy and 49.5 FPS** on mobile, trained on a 4,594-sample French Sign Language dataset built from scratch with a 6-person team.

Additional — 2023 – 2026

- **LangDetect** — 98.29% accuracy across 28 languages (TF-IDF n-grams + LinearSVC). **RAG Document Assistant** — LlamaIndex, Qdrant, Ollama with source citations. **Robotic Arm RL** — PPO, 36 parallel agents, 70% success, emergent inverse kinematics. **AI Resume Screener** — n8n + Gemini, 95% less screening time. **SchemaForge** — natural language to ER diagrams. **Darija Chatbot** — 183K-parameter LSTM. **vSlide** — solo-published Unity game, 9+ months on Google Play.

EDUCATION

Master of Excellence, Artificial Intelligence and Connected Objects — *Ibn Tofail University* — 2024 – 2026

Bachelor of Excellence, AI and Data Engineering — *Ibn Tofail University* (Good) — 2023 – 2024

SKILLS

Languages: Python, TypeScript, JavaScript, C#, SQL, C++

ML & Vision: PyTorch, TensorFlow, Transformers, Hugging Face, Scikit-learn, ONNX, YOLO, RT-DETR, CNN and transformer architectures (PP-LCNet, U-Net, ResNet, ViT), MediaPipe, OpenCV, model training from scratch, benchmarking and evaluation

Document AI & NLP: Docling, PaddleOCR, Tesseract, layout detection, table-structure recognition, RAG, LlamaIndex, LangChain, llama.cpp, Ollama, Qdrant, constrained decoding, prompt engineering, AI agents, n8n

Engineering: FastAPI, Flask, Supabase, PostgreSQL, React, React Native, Expo, Tailwind, Docker, Git, Linux, Unity ML-Agents, security hardening

LANGUAGES

Arabic (native) · English (C1) · French (B2)